From Statistical Features to Contextual Embeddings: A Multi-Paradigm Evaluation of Emotion Classification Models

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Abstract

Emotion classification has evolved through three distinct representational paradigms statistical features, static word embeddings, and contextual transformer representations yet the practical performance boundaries between them remain poorly understood. This study evaluates 14 model variants across all three paradigms on a six-class sentiment dataset, revealing that the transition from sparse statistical features to pre-trained GloVe embeddings accounts for the largest performance gain (+3.6 points F1), while full transformer fine-tuning with BERT and RoBERTa yields only a marginal additional improvement (+0.8 points) at substantially higher computational cost. LinearSVC with TF-IDF emerges as a surprisingly strong baseline at 89.3% F1, Bidirectional GRU with GloVe achieves 92.9% at a fraction of transformer overhead, and BERT/RoBERTa reach a ceiling of 93.7%. These results suggest that for resource-constrained deployment, embedding-based recurrent architectures offer a near-optimal balance between performance and efficiency that transformer models rarely justify in practice.

Keywords: emotion classification, GloVe, FastText, BERT, RoBERTa, transformer fine-tuning, natural language processing

Code: The implementation for all experiments in this paper is publicly available at `https://github.com/GregReynaldi/emotion-classification-benchmark`

1 Introduction

Emotion classification remains one of the most actively studied problems in natural language processing, driven by the growing volume of opinionated text across social media, product platforms, and news commentary. Despite decades of research, practitioners still face a non-trivial model selection problem: the landscape ranges from lightweight statistical classifiers trainable in seconds to large pre-trained language models requiring significant GPU resources, and the empirical trade-offs between them are rarely characterized in a unified, controlled setting.

The field has progressed through three recognizable phases. Early approaches relied on sparse, frequency-based text representations such as Bag-of-Words and TF-IDF, paired with discriminative classifiers like Naive Bayes, Logistic Regression, and Support Vector Machines. These methods proved surprisingly effective and remain competitive baselines. The second phase introduced dense word embeddings, notably GloVe and FastText, which encode distributional semantics learned from large corpora and enabled recurrent architectures such as Bidirectional LSTM and GRU to model sequential context more effectively. The third and current phase is dominated by transformer-based pre-trained language models including BERT, RoBERTa, and DistilBERT, which produce deeply contextualized token representations and have set state-of-the-art benchmarks across most NLP tasks.

What remains underexplored is a direct, side-by-side comparison of all three paradigms under identical experimental conditions. Most published studies evaluate within a single paradigm, making it difficult to assess where the meaningful performance boundaries lie or whether the computational cost of transformers is justified for a given task. This paper addresses that gap directly. All model variants spanning all three families are evaluated on a consistent six-class sentiment dataset, with results analyzed not only by accuracy and F1-score but also by training time and model size, factors that matter considerably in real-world deployment.

2 Related Work

2.1 Classical Machine Learning for Text Classification

Early text classification systems were built almost entirely on sparse, count-based representations. The Bag-of-Words model treated documents as unordered collections of term frequencies, while TF-IDF refined this by down-weighting terms that appear frequently across all documents and therefore carry little discriminative value. Paired with Naive Bayes, these representations produced surprisingly competitive classifiers given their simplicity, largely because the conditional independence assumption aligns well with high-dimensional sparse inputs [1]. Support Vector Machines later became the dominant choice for text tasks, benefiting from their ability to find maximum-margin decision boundaries in spaces where the number of features often exceeds the number of training samples [2]. N-gram extensions provided modest gains by capturing short local phrase patterns, and gradient boosting methods such as XGBoost offered further improvements at the cost of substantially longer training times [3, 4]. In this study, these methods serve as the first paradigm baseline. The results confirm that LinearSVC with TF-IDF remains a strong and efficient classifier, reaching 89.3% F1-score with sub-second training time, a finding that challenges the assumption that classical models are no longer worth considering.

2.2 Word Embedding-Based Deep Learning

The shift from sparse statistical features to dense word embeddings represented the first major leap in representational capacity for NLP. Word2Vec showed that embeddings trained on raw text encode semantic and syntactic structure in their geometry [5]. GloVe refined this by incorporating global co-occurrence statistics, while FastText extended coverage to morphologically complex and out-of-vocabulary words through subword representations [6, 7]. When these embeddings are used as inputs to Bidirectional GRU or LSTM networks, the model gains the ability to process word meaning and sequential context simultaneously, something that bag-of-words approaches cannot do by design [8, 9, 10]. In the experiments reported here, a Bidirectional GRU with GloVe embeddings reaches 92.9% F1-score, representing a 3.6-point improvement over the best classical baseline. This gap is the largest single performance jump observed across all three paradigms evaluated in this work, which suggests that the transition from statistical features to pre-trained embeddings is where the most meaningful representational gain occurs.

2.3 Transformer-Based Pre-Trained Language Models

The Transformer architecture introduced self-attention as the primary mechanism for modeling relationships between tokens, replacing the sequential inductive bias of recurrent networks with a fully parallel computation over the entire input sequence [11]. BERT operationalized this for transfer learning by pre-training a deep bidirectional model on masked language modeling, then fine-tuning on downstream tasks with minimal architectural changes [12]. RoBERTa improved on BERT by training longer on more data with dynamic masking and without the next sentence prediction objective, consistently yielding higher benchmark scores [13]. DistilBERT applied knowledge distillation to produce a smaller, faster model that retains most of BERT's performance at roughly 40% fewer parameters [14]. In this study, BERT and RoBERTa both reach 93.7% F1-score, which is the performance ceiling across all evaluated models. However, the margin over the best embedding-based model is only 0.8 points, raising a practical question that this paper directly addresses: whether that marginal gain justifies the substantially higher computational cost in resource-constrained settings.

3 Dataset and Preprocessing

3.1 Dataset Description

All experiments use the Emotion Dataset from Kaggle [15], consisting of 20,000 short English text samples labeled across six emotional categories: anger, fear, joy, love, sadness, and surprise. The dataset reflects naturalistic emotional language drawn primarily from social media. Despite covering six distinct emotions, several categories share considerable vocabulary overlap, most notably joy and love, and anger and fear, making this a genuinely challenging multi-class problem. The data is split using a stratified 80/20 ratio, yielding 16,000 training samples and a fixed 4,000-sample test set held constant across all 14 model variants to ensure fair comparison.

3.2 Preprocessing Pipeline

A single preprocessing pipeline is applied consistently across all model families. The core steps are removing special characters and punctuation, collapsing multiple whitespace characters into single spaces, and lowercasing all tokens.

For classical ML models, the cleaned text is vectorized using CountVectorizer, TF-IDF, and TF-IDF with bigram features, each with a minimum document frequency of 5. For embedding-based models, tokens are mapped to 300-dimensional vectors using GloVe (Wikipedia 2014 + Gigaword 5, 6B tokens) and FastText (Wikipedia 2017 + UMBC + statmt.org, 16B tokens). Missing tokens receive zero vectors and all sequences are padded or truncated to 100 tokens. When both embeddings are combined, vectors are concatenated to produce a 600-dimensional input. For transformer models, raw unprocessed text is passed directly to each model's native tokenizer, WordPiece for BERT and DistilBERT and BPE for RoBERTa, with a maximum sequence length of 100 tokens.

4 Methodology

4.1 Baseline Machine Learning Models

The baseline experiments cover four classical classifiers: Naive Bayes, Logistic Regression, LinearSVC, and XGBoost. Each is evaluated against three feature extraction configurations, namely CountVectorizer, TF-IDF, and TF-IDF with bigram features, producing twelve model variants in total. A minimum document frequency of 5 is applied across all vectorizers to filter terms that appear too rarely to be informative. An additional MLP baseline is included, built as a three-layer dense network with ReLU activations and dropout regularization, trained on TF-IDF features. Class weight balancing is applied to all models given the uneven distribution across the six emotion categories. All experiments use scikit-learn defaults with random seed fixed at 42.

4.2 Bidirectional GRU and LSTM with Pre-Trained Embeddings

The embedding-based experiments center on a stacked Bidirectional GRU architecture built with the Keras Sequential API. The first layer uses 128 units and the second uses 64 units, both with return_sequences enabled to retain the full temporal output. Rather than using a single pooling strategy, global average pooling and global max pooling are computed separately across the time dimension and concatenated, which in practice captures both the overall distributional tendency and the most salient features within the sequence. A dropout layer with rate 0.2 is placed before the final six-class softmax output.

Three embedding configurations are tested to isolate the contribution of each embedding source: GloVe alone (300 dimensions), FastText alone (300 dimensions), and a combined input formed by concatenating both vectors per token (600 dimensions). A Bidirectional LSTM with the same architecture and combined embeddings is included to assess whether the more complex gating mechanism of LSTM offers any practical advantage over GRU on this task. All models are trained with Adam at a learning rate of 5e-4, batch size 64, up to 100 epochs, with early stopping on validation loss at patience 30. The architecture is illustrated in Figure 1.

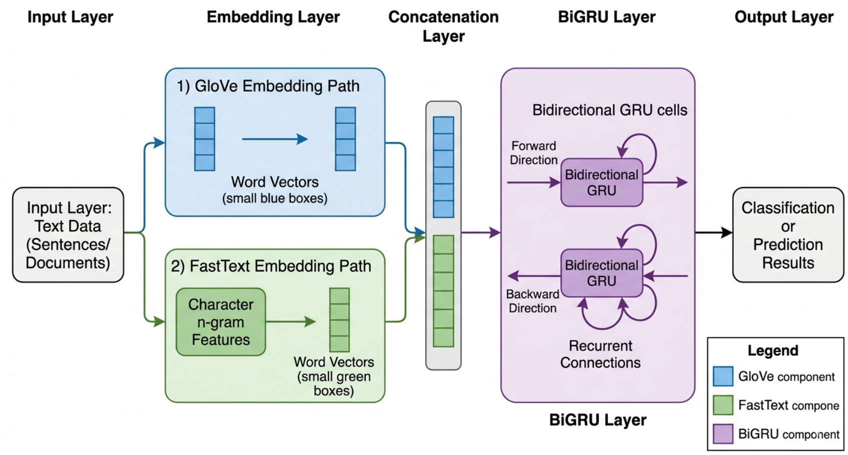

Figure 1: BiGRU using GloVe and FastText

4.3 Transformer Fine-Tuning

Three transformer models are fine-tuned for the six-class emotion classification task: DistilBERT (distilbert-base-uncased, 66.96M parameters), BERT (bert-base-uncased, 109.49M parameters), and RoBERTa (roberta-base, 124.65M parameters). Each model is loaded from its Hugging Face checkpoint with a classification head attached in place of the language modeling head. Fine-tuning follows a shared configuration across all three: learning rate 2e-5, weight decay 0.01, batch size 16 for training and 8 for evaluation, up to 10 epochs, with mixed precision training enabled and best checkpoint selection based on weighted F1-score. The choice to evaluate all three models under identical conditions is deliberate, as it isolates the effect of model architecture and pre-training strategy rather than tuning differences.

5 Results and Discussion

5.1 Baseline Machine Learning Models

Table 1 reports performance across all classical model and vectorizer combinations. LinearSVC with TF-IDF stands out as the strongest baseline, reaching 89.3% accuracy and 89.2% F1-score while training in under one second. This result is worth noting because it demonstrates that a well-configured classical model remains a serious option, not merely a reference point.

One finding that deserves attention is the behavior of Naive Bayes under different vectorizers. With CountVectorizer it reaches 83.5% accuracy, but drops to 72.9% when switched to TF-IDF. This is consistent with the known theoretical mismatch between the Naive Bayes independence assumption and normalized TF-IDF weights, and it serves as a practical reminder that vectorizer choice is not interchangeable across classifiers. XGBoost performs competitively at 87.2 to 88.4% across configurations, but its training time with bigram TF-IDF reaches 37.33 seconds, which is notably high for a baseline model with no meaningful accuracy advantage over LinearSVC.

5.2 Bidirectional GRU and LSTM Models

Table 2 shows that moving from statistical features to pre-trained embeddings produces the single largest performance jump observed in this study. The best embedding-based model, Bidirectional GRU with combined GloVe and FastText, reaches 93.2% accuracy, which is 3.9 points higher than the best classical baseline. This gap is more significant than the one between embedding models and transformers, which suggests that the representational shift from sparse features to dense embeddings is where the most fundamental improvement in understanding emotional language occurs.

GloVe and FastText capture different but complementary aspects of lexical meaning. GloVe constructs its representations by factorizing a global word co-occurrence matrix, meaning each vector encodes

Table 1: Performance of Baseline Machine Learning Models

Model	Vectorizer	Accuracy	Precision	Recall	F1-Score	Train Time (s)
Naive Bayes	CountVectorizer	0.835	0.832	0.835	0.830	2.64
Naive Bayes	TF-IDF	0.729	0.787	0.729	0.682	0.12
Naive Bayes	TF-IDF + NGram	0.685	0.769	0.685	0.624	0.58
Logistic Regression	CountVectorizer	0.885	0.884	0.885	0.884	3.19
Logistic Regression	TF-IDF	0.866	0.868	0.866	0.861	3.23
Logistic Regression	TF-IDF + NGram	0.834	0.843	0.834	0.825	8.05
LinearSVC	CountVectorizer	0.880	0.880	0.880	0.880	1.30
LinearSVC	TF-IDF	**0.893**	**0.892**	**0.893**	**0.892**	0.55
LinearSVC	TF-IDF + NGram	0.887	0.886	0.887	0.886	1.49
XGBoost	CountVectorizer	0.884	0.887	0.884	0.885	3.53
XGBoost	TF-IDF	0.872	0.874	0.872	0.873	12.31
XGBoost	TF-IDF + NGram	0.876	0.878	0.876	0.877	37.33
MLP	TF-IDF	0.828	0.836	0.828	0.830	–
MLP	CountVectorizer	0.853	0.855	0.853	0.853	–
MLP	TF-IDF + NGram	0.840	0.841	0.840	0.840	–

how a word relates to every other word across the entire corpus. This gives GloVe strong sensitivity to semantic context, which is particularly useful for emotion classification where words like "devastated" or "thrilled" carry meaning largely through the company they keep in text. FastText, by contrast, operates at the subword level by decomposing each word into character n-grams before learning representations. This makes it more robust to informal spelling, morphological variation, and words that fall outside the training vocabulary, all of which are common in social media text. When the two are concatenated into a single 600-dimensional input, the model has access to both corpus-level semantic relationships and subword-level lexical structure simultaneously.

Within this model family, the GloVe-only Bidirectional GRU is particularly noteworthy. At 92.9% accuracy, it trails the combined embedding variant by only 0.3 points, yet trains faster and produces a model that is 1.74 MB compared to 2.62 MB. This suggests that for this dataset, the semantic coverage provided by GloVe alone is sufficient to capture most of the emotionally relevant signal, and the subword robustness of FastText contributes only marginally. Adding FastText to GloVe costs more than it contributes. The Bidirectional LSTM with combined embeddings tells a similar story: it underperforms the GRU counterpart by 1.2 points while taking nearly 1.8 times longer to train, which in practical terms means a more complex architecture delivered worse results at greater cost.

Table 2: Performance of Embedding-Based Models

Model	Embeddings	Accuracy	F1-Score	Model Size
BiGRU	GloVe + FastText	**0.932**	**0.932**	2.62 MB
BiGRU	GloVe Only	0.929	0.929	1.74 MB
BiGRU	FastText Only	0.925	0.927	1.74 MB
BiLSTM	GloVe + FastText	0.920	0.921	**3.48 MB**

5.3 Transformer-Based Models

Table 3 shows that BERT and RoBERTa both reach 93.6% accuracy and 93.7% F1-score, representing the performance ceiling across all models evaluated in this study. RoBERTa achieves slightly higher precision at 94.0% compared to BERT at 93.8%, which likely reflects the impact of its more rigorous pre-training setup rather than any architectural difference. DistilBERT reaches 93.2% accuracy with 66.96M parameters, roughly 39% fewer than BERT, and matches the best embedding-based model exactly. For use cases where model size matters, DistilBERT offers a compelling middle ground between the efficiency of recurrent models and the performance of full transformers.

Table 3: Performance of Transformer-Based Models

Model	Accuracy	Precision	Recall	F1-Score	Parameters
DistilBERT	0.932	0.934	0.932	0.932	66.96M
BERT	**0.936**	0.938	**0.936**	**0.937**	109.49M
RoBERTa	**0.936**	**0.940**	**0.936**	**0.937**	**124.65M**

5.4 Cross-Paradigm Comparison and Efficiency Analysis

Table 4 brings together the top-performing model from each paradigm. The overall picture is clear: each successive paradigm improves on the previous one, but the size of that improvement shrinks considerably as model complexity grows. Going from classical ML to pre-trained embeddings adds 3.6 accuracy points. Going from embeddings to transformers adds only 0.7 points further.

What makes this finding practically meaningful is the cost attached to that 0.7-point gain. Classical models train in seconds. Embedding-based models take 15 to 31 minutes. Transformer fine-tuning requires GPU resources and significantly longer training cycles. For teams building emotion classification systems under real constraints, the Bidirectional GRU with GloVe embeddings sits at a point on the performance-efficiency curve that is difficult to argue against: 92.9% accuracy, compact model size, and no dependency on large-scale pre-trained language models.

Table 4: Cross-Paradigm Performance Comparison (Top Models per Family)

Rank	Model	Paradigm	Accuracy	F1-Score
1	**BERT**	**Transformer**	**0.936**	**0.937**
1	**RoBERTa**	**Transformer**	**0.936**	**0.937**
3	BiGRU (GloVe+FT)	Embedding-based	0.932	0.932
3	DistilBERT	Transformer	0.932	0.932
5	BiGRU (GloVe)	Embedding-based	0.929	0.929
6	BiGRU (FastText)	Embedding-based	0.925	0.927
7	BiLSTM (GloVe+FT)	Embedding-based	0.920	0.921
8	LinearSVC (TF-IDF)	Classical ML	0.893	0.892
9	LogReg (CountVec)	Classical ML	0.885	0.884
10	XGBoost (CountVec)	Classical ML	0.884	0.885

6 Conclusion

This study set out to answer a practical question: across three generations of text representation, where do the meaningful performance boundaries actually lie for emotion classification? The results are clearer than expected.

The most important finding is not that transformers perform best. It is that the largest single gain in this entire study comes from switching away from statistical features entirely. Moving from LinearSVC with TF-IDF to a Bidirectional GRU with GloVe embeddings adds 3.6 accuracy points, which accounts for the vast majority of the total improvement achievable. The further step to BERT or RoBERTa adds only 0.7 points more, at a cost in compute and model size that is orders of magnitude higher.

The reason transformers still hold the performance ceiling comes down to how they process language. Unlike GRU and LSTM networks that read text sequentially and compress earlier context into a fixed hidden state, the self-attention mechanism in BERT and RoBERTa allows every token to directly relate to every other token in the sequence simultaneously. For emotion classification this matters because emotional meaning is frequently non-local. A negation word early in a sentence can completely reverse the tone of what follows, and self-attention captures this naturally. Beyond architecture, these models also arrive at fine-tuning pre-trained on billions of tokens, carrying a depth of linguistic knowledge that a recurrent model trained on 16,000 samples simply cannot replicate. The 0.7-point margin over GloVe-based models is modest in percentage terms, but it reflects a qualitatively different kind of language understanding rather than just a better version of the same approach.

For most real deployment scenarios however, the Bidirectional GRU with GloVe-only embeddings remains a more practical recommendation: 92.9% accuracy, 1.74 MB model size, no GPU requirement for inference, and training that completes in under 15 minutes. Adding FastText on top of GloVe contributes only 0.3 points at the cost of larger model size and longer training. Replacing GRU with LSTM produced worse results at nearly double the training time, which suggests that architectural complexity alone does not translate to better emotion understanding. For teams who need something even leaner, LinearSVC with TF-IDF at 89.3% F1-score trains in under a second with no deep learning infrastructure required at all.

Future work will explore lightweight transformer variants such as TinyBERT and MobileBERT to examine whether the efficiency gap can be narrowed further, cross-lingual transfer for emotion classification in low-resource languages, and ensemble approaches combining statistical and neural representations to assess whether complementary strengths across paradigms can be leveraged jointly.

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